

Varun Kannan

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Website: <https://www.qgspinor.com>

Education

2018-09 - 2022-06: Bachelor of Science: Physics, Minor in Mathematics

University of California - San Diego - La Jolla, CA

GPA: 3.32 | 3.60 Major

Honors: Provost Honors (5 terms), Salutatorian (HS)

Research Interests

Quantum Gravity | Quantum Information | Quantum Technology

Technical Skills

Programming Languages: Python, Java, Mathematica, C#, JavaScript, HTML

Programming related frameworks: Pandas, NumPy, Matplotlib/Plotly, VectorBt, OOP, Multiprocessing

Programming related platforms: Linux, WSL, PowerShell, Jupyter Lab, Software Related IDEs

Research Experience

Quantum Physics Research

Internship (Paid): 06/2021 - 09/2021 (3 months) UCSD Physics Department, La Jolla, CA

Dr. Daniel Green, Associate Professor at UCSD

- Identified issues in existing models of quantum fields and tested proposed models on existing data with rigorous analysis tools and a comprehensive research process. Use of Bayesian filters to estimate hidden variables in the context of quantum phenomena, such as decoherence.
- Use of Axiomatic QFT and statistical learning principles to ensure both theoretical and modeling consistency
- Analyzed research data from various technical sources using Mathematica to create representative graphs and summaries highlighting key insights.

Honors Research: 09/2021 - 12/2021 (3 months)

- Continued the previous internship as an official research subproject under Dr. Green

Independent Study/Research 03/2021 - 12/2021 (9 months) UCSD Physics Department, La Jolla, CA

Dr. George Fuller, Distinguished Professor, Former Director of the Center for Astrophysics and Space Science, UCSD

- A mentorship and research on various topics in the field of General Relativity, including the formulation of alternate descriptions of gravity as a geometrical theory. Obtained testable predictions and attempted to compare to existing data.
- Studied extensions to the torsion free model, including the use of the spin connection, and the use of a proposed operator theoretic construction of the related geometric quantities.

Quantum Gravity Study 08/2024 - 06/2025 (10 months) UCSD Mathematics/Physics Department, La Jolla, CA

- A study of the description of gravity as a gauge theory built on geometric principles. Proposed theoretical models and studied existing ones. Focused on the use of non-Abelian gauges to represent relativistic phenomena. Authored a paper which summarized my research.
- Reviewed/Overseen by Dr. Jeffrey Rabin (UCSD Mathematics Professor)

Quantitative Computational Research 06/2025 - Current (11 months):

- Python-based backtesting framework built for research purposes and general usability (GitHub: <https://github.com/sponge-mi-boi/Quant-data>)
- Designing Trading Strategies with accurate signal extraction, model validation, and scalability analysis on low/medium frequency timescales
- Evaluated with the use of risk-adjusted performance metrics
- Vectorized research pipeline focused on efficient data extraction and the use of walk-forward analysis to validate both out of sample performance and multi-regime stability
- Testing of various quantitative methods including PCA-based factor models, multi-stock cointegration, HMMs, and statistical arbitrage based portfolios
- Results/methodology is being documented publicly: [Quant Finance](https://www.qgspinor.com/writings_learnings/writings)

Publications and Technical Writing

- Authored a paper based on individual study along with a review from Dr. Jeffrey Rabin on the application of the geometric interpretation of gauge theories to formulating models in quantum gravity
https://www.qgspinor.com/writings_learnings/writings

- Created the website, <https://www.qgspinor.com>, where I post technical writing, short review articles, derivations, and related exploratory topics.

Awards

- Provost Honors: Fall 2019, Fall 2020- Spring 2021, Spring 2022
- Completion of a quarter of Honors Research under Dr. Daniel Green

References

References available upon request